



Research Paper

Multiphysics modeling of UO_2 -SiC composite fuel performance with enhanced thermal and mechanical properties



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HIGHLIGHTS

- LWR fuel performance code (CAMPUS) developed.
- UO_2 -SiC fuel properties and physical phenomena modeled and implemented.
- Significant decrease in fuel temperature and FGR for UO_2 -SiC fuel achieved.
- Delay in pellet cladding interaction for UO_2 -SiC fuel demonstrated.
- Reactor safety for UO_2 -SiC fuel improved.

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ABSTRACT

The relatively poor thermal conductivity of the UO_2 fuel is a major challenge for optimizing reactor operation and safety performance, despite its widespread use in the majority of power reactors. The already low thermal conductivity further degrades with burnup due to accumulation of defects, fission product precipitates and fission gas bubbles. The high thermal stresses cause significant pellet cracking, leading to more pellet expansion and causing pellet cladding interaction and the release of fission product gases. Mitigation of these phenomena is accomplished by limiting operating power and ramp rates at high burnup. This study presents the development of a model for UO_2 -SiC composite light water reactor fuel with enhanced thermal and mechanical properties to mitigate some of the material performance limitations. The model is constructed using self-defined physics modules which are fully coupled and solved using the COMSOL Multiphysics platform. Key fuel performance phenomena being considered include heat generation and conduction, species diffusion, thermomechanics (thermal expansion, elastic strain, densification, and fission product swelling strain), grain growth, fission gas production and release, gap heat transfer, mechanical contact, gap/plenum pressure with plenum volume, cladding thermal and irradiation creep and oxidation. All the equations are solved via finite-element method on a 2D axisymmetric geometry of a fuel pellet and cladding. The use of UO_2 -SiC will enhance thermal conductivity, which in turn will decrease nuclear fuel temperatures. This subsequently will improve fission gas retention and reduce pellet cladding interaction resulting from fuel cracking, relocation, and swelling.

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1. Introduction

The vast majority of operating commercial power reactors are fueled with sintered Uranium Dioxide (UO_2) pellets. This is the fuel of choice for most reactor designs including PWR, BWR, PHWRs, and gas and metal cooled reactors. This popularity is a result of several very desirable chemical and physical properties such as:

high melting point, good high-temperature stability, good chemical compatibility with cladding and coolant, reasonable resistance to radiation and fission product swelling and adequate uranium density. However, a major disadvantage is the low thermal conductivity of UO_2 , which causes both a high centerline temperature for the fuel pellets and a large radial temperature gradient. When the thermal expansion is considered, the high temperatures and temperature gradients contribute to fuel cracking, pellet cladding interaction and the release of fission product gases as demonstrated in Refs. [1–4].

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Nomenclature

bu	burnup [%]	p_0	fuel initial porosity fraction
Bu	burnup [fissions/atoms-U]	Q	heat generation rate [W/m ³]
C	atomic fraction of oxygen interstitials [atoms/m ³]	Q_{fg}	volumetric rate of production of fission gas atoms [atoms/(m ³ s)]
C	concentration of fission gas atoms [atoms/m ³]	Q_i	activation energies [J/mol]
C_p	specific heat [J/(kg K)]	Q_r	net radial heat flux from cladding to bulk coolant [W/m ²]
C_r	roughness coefficient	Q_{oxide}	heat flux reduction due to cladding oxide layer growth [W/m ²]
C_s	constant	Q_1	activation energy for pre-transition oxidation [J/mol]
C_1	rate constant for pre-transition oxidation [m/s]	Q_2	activation energy for post-transition oxidation [J/mol]
C_2	rate constant for post-transition oxidation [m/s]	R	universal gas constant [J/(mol K)]
D	diffusion coefficient of fission gas [m ² /s]	r	surface roughness [m]
D_f	fuel density in percent of theoretical one	r_f	grain boundary bubble radius [m]
d_g	gap width [m]	R_{gb}	fission gas release rate to grain boundary [atoms/(m ³ m s)]
D_O	diffusion coefficient of oxygen [m ² /s]	T	temperature [K]
d_{pen}	penetration distance [m]	T_n	$T_n = T(K)/1000$
E	Young's modulus [Pa]	t	time [s]
E_D	activation energy for Frenkel defects [J/mol]	t_{oxide}	oxide layer thickness [m]
F	thermodynamic factor of oxygen	t_{trans}	oxide layer thickness at transition period [m]
$\dot{F}$	volumetric fission rate [fissions/(m ³ s)]	$\mathbf{u}$	displacement vector [m]
f_B	grain boundary fraction at saturation	V	volumetric fraction
f_d	dissolved fission products correction	V	volume [m ³]
f_p	precipitated fission products correction	v	phonon velocity [m/s]
f_{por}	porosity correction	v_l	longitudinal phonon velocity [m/s]
f_r	radiation damage correction	v_t	transverse phonon velocity [m/s]
f_x	deviation from stoichiometry	W	weight fraction
F_{dens}	fraction of fuel porosity removed by densification	Y	oxygen to metal ratio
F_v	body force per unit volume [N/m ³]	X_{dev}	oxygen to metal ratio deviation
g	surface jump distance [m]	α	coefficient of thermal expansion
G	grain size [μ m]	α	radius of particle [μ m]
G	shear modulus [Pa]	a_{pellet}	fuel pellet radius [m]
G_{bsat}	saturation fission gas release of grain boundary [atoms/m ³]	δ	average gas film thickness [m]
g_d	average fuel grain diameter [m]	σ	Stephan-Boltzman constant [kg/(s ³ K ⁴)]
g_r	local fuel grain radius [m]	σ	Cauchy stress tensor [Pa]
g_{max}	limiting grain size [m]	σ	effective (Mises) stress [Pa]
h	conductance [W/(m ² K)]	ε	radiating surface emissivity
h_c	interfacial thermal conductance [W/(m ² K)]	ε	elastic strain vector
H	Meyer hardness of the softer material [Pa]	$\dot{\varepsilon}$	creep rate [1/s]
h_{film}	cladding effective heat transfer coefficient [W/(m ² K)]	$\dot{\varepsilon}_{ir}$	effective irradiation creep rate [1/s]
J	oxygen flux [atoms/(m ² s)]	$\dot{\varepsilon}_{ss}$	effective thermal creep rate [1/s]
k	rate constant [m ² /s]	ε_D	fraction of volume reduction due to fuel densification
k	thermal conductivity [W/(m K)]	ρ	density [kg/m ³]
k_{95}	thermal conductivity of 95%TD fuel [W/(m K)]	θ	the Einstein temperature [K]
k_B	Boltzmann constant [m ² /(s ² K)]	θ_{fg}	semi-dihedral angle between bubble surfaces [°]
L	length [m]	ν	Poisson's ratio
m	matrix	Φ	fast neutron flux [n/(m ² s)]
k_{eff}	effective thermal conductivity [W/(m K)]	γ_{se}	surface tension of the bubbles [J/m ²]
n	moles of gas [mol]	$\Delta \varepsilon_{sw-s}$	volumetric solid swelling increment [m ³]
p	fuel porosity fraction	$\Delta \varepsilon_{sw-g}$	volumetric gas swelling increment [m ³]
p	particle	FEA	finite element analysis
P	pressure [Pa]	FGR	fission gas release
P_c	contact pressure [Pa]	LWR	light water reactor
P_{est}	estimated contact pressure [Pa]	TD	theoretical density
P_{ext}	externally applied hydrostatic pressure [Pa]		
P_f	penalty factor [Pa/m]		
P_i	interface pressure [Pa]		
P_{lin}	linear power [W/m]		

In principle, the high fuel centerline temperature and temperature gradients can be decreased and the reactor performance improved by developing a higher thermal conductivity variant of UO₂ fuel. Such a fuel would experience a reduction in pellet cladding interaction through lower thermal expansion and lower thermal stresses as described in [1,3]. Additionally, lower

temperatures reduce fission product mobility thereby reducing fission gas release, grain boundary swelling, and stress-corrosion cracking. This would enable higher fuel burn-up, and improved reactor safety due to faster thermal response, lower grain boundary inventory of fission products and less thermal energy in the fuel pins.

The improved thermal conductivity could be achieved by producing a composite material with UO_2 and a suitable high thermal conductivity material with appropriate chemical and physical properties. A large number of materials have been proposed including carbon (in the form of graphite, diamond and nanotubes) [5,6], ceramics (such as ThO_2 , SiC, UN and BeO) [7–10], and some high temperature metals in various configurations such as homogenous or heterogeneous materials or heat-pipes [11]. Ceramic-ceramic composites are advantageous for their high melting point and chemical stability.

Mixtures and composites of UO_2 and ThO_2 have been widely studied for many decades in part because the thorium is a fertile material. However, the thermal conductivity of thorium-dioxide is only slightly higher than that of UO_2 , and therefore requires large volume fractions in order to observe a meaningful improvement. In fact, some homogenous mixtures of (U, Th) O_2 can actually have lower thermal conductivity compared to pure UO_2 , depending on the composition [7].

Silicon Carbide and Beryllium Oxide are two of the most promising materials that demonstrate compatibility with UO_2 , and have high thermal conductivities but have received less attention [12,13]. The enhanced thermal conductivity UO_2 -BeO composite fuel performance has been studied recently by Liu et al. [14] through the development of computer models with augmented material properties. The model predicts significant reduction in fuel temperature, fission gas release and cladding strain, including a roughly 370 K reduction in fuel centerline temperature and a 66.7% reduction in fission gas release from UO_2 -13.6 wt%BeO fuel case at fuel burnup of 600 MW h/kg-U, thereby improving fuel margins. In 2013, a series of UO_2 -SiC composite fuel pellets with different sizes and volumetric fractions of SiC particles was fabricated by Yeo [8], and the thermal conductivities of the composite fuels were measured and compared to the results determined by theoretical formulations available in the literature. The in reactor fuel performance of the UO_2 -SiC with enhanced thermal conductivity and mechanical properties has not been experimentally reported or theoretically simulated. In this paper, theoretical (modeling) studies were performed to quantify the impact of UO_2 -SiC composites for a range of irradiation conditions.

Understanding and predicting the evolving behaviors of nuclear fuels are major challenges for the analysis, design, optimization, and safe operation of new fuel types. The multiphysics behavior of nuclear fuels involves tightly coupled thermal, mechanical and chemical phenomenon associated with a large number of control parameters and uncertainties, which requires advanced fuel performance modeling and simulation capability.

In this paper, a CityU Advanced Multiphysics Nuclear Fuels Performance with User-defined Simulations (CAMPUS) code has been developed to simulate UO_2 -SiC composite fuel performance. The CAMPUS code is based on the multiphysics, multidimensional, multipurpose numerical engine approach used in the BISON code from Idaho National Laboratory [15,16] and the FAST code developed at Royal Military College of Canada and Canadian Nuclear Laboratories [17,18]. The CAMPUS code is based on the framework of COMSOL Multiphysics and uses user-defined multiple physical models. COMSOL Multiphysics (COMSOL, Inc., 2015) is a general purpose finite element analysis solver and simulation software/FEA software package for various mathematics, physics and engineering applications, especially useful for coupled nonlinear phenomena. COMSOL can be controlled via the Graphical User Interface (GUI) or the Application Programming Interface (API) scripting via JAVA or MATLAB. The CAMPUS modeling and simulation code structure is shown in Fig. 1. In the approach, the phenomena, geometry and materials properties were implemented within the code. The resulting system of equations is nonlinear, strongly

coupled and includes integral, differential and algebraic equations. These are discretized and solved numerically using the COMSOL engine.

The resulting CAMPUS model was then used to predict UO_2 -SiC fuel performance with different volumetric fractions of SiC under light water reactor irradiation conditions. The simulation geometry includes fuel pellet, gap, plenum and cladding. The materials properties are semi-empirical correlations which dependent on temperature, burnup, fission rate, grain size and oxygen-to-metal ratio. Through fully coupled Multiphysics modeling, a more complete and quantitative understanding of the benefits of UO_2 -SiC fuel performance in a light water reactor is demonstrated in this paper.

2. Material properties

2.1. UO_2 -SiC fuel properties

2.1.1. Thermal conductivity

In general, the UO_2 -SiC fuel thermal conductivity is given as in Eq. (1) accounting for the effects of dissolved fission products, precipitated fission products, porosity, deviation from stoichiometry, and radiation damage, as for UO_2 fuel [19,20].

$$k = k(\text{UO}_2\text{-SiC}) \cdot f_d \cdot f_p \cdot f_{por} \cdot f_x \cdot f_r \quad (1)$$

where f_d – dissolved fission products correction; f_p – precipitated fission products correction; f_{por} – porosity correction; f_x – deviation from stoichiometry; f_r – radiation damage correction.

The as-fabricated UO_2 -SiC fuel thermal conductivity accounting for the porosity of the fuel is calculated by multiplying a factor to 95% theoretical density UO_2 -SiC fuel thermal conductivity as shown in Eq. (2) [20]:

$$k(\text{UO}_2\text{-SiC}) = k_{95}(\text{UO}_2\text{-SiC}) \cdot \left(\frac{1}{1 - (2.6 - 5 \times 10^{-4}T) \cdot 0.05} \right) \quad (2)$$

Yeo [8] fabricated samples of UO_2 -SiC composite fuel using the Spark Plasma Sintering (SPS) technique, and compared three models for predicting the effective thermal conductivity of the composite materials in his thesis [8], namely: the Maxwell model, the Hasselman and Johnson Model, and the Nan's effective thermal conductivity model. The Hasselman and Johnson model was adopted in this work as it was found to produce good agreement with the measured thermal conductivity [21]. Using this model, the effective thermal conductivity of composites is expressed as

$$k_{eff} = k_m \frac{2 \left(\frac{k_p}{k_m} - \frac{k_p}{\alpha h_c} - 1 \right) V_p + \frac{k_p}{k_m} + 2 \frac{k_p}{\alpha h_c} + 2}{\left(1 - \frac{k_p}{k_m} + \frac{k_p}{\alpha h_c} \right) V_p + \frac{k_p}{k_m} + 2 \frac{k_p}{\alpha h_c} + 2} \quad (3)$$

where k_{eff} is the effective thermal conductivity, and subscripts p and m are particle and matrix, respectively, V_p is the volumetric fraction of particles, α is the radius of particle and h_c is the interfacial thermal conductance. The interfacial conductance can be estimated using the acoustic mismatch model of Swartz and Pohl [22], where the interfacial thermal conductance is given by,

$$h_c \approx \frac{1}{2} \rho_m \cdot C_p \cdot \frac{v_m^3}{v_p^2} \cdot \frac{\rho_m \rho_p v_m v_p}{(\rho_m v_m + \rho_p v_p)^2} \quad (4)$$

where ρ is density, C_p is the specific heat capacity of matrix, v is phonon velocity, and subscripts p and m refer to particle and matrix, respectively. The phonon velocities of UO_2 matrix and SiC particle can be estimated using the following equation [23].

$$\frac{1}{v_l^2} + \frac{2}{v_t^2} = \frac{3}{v^2} \quad (5)$$

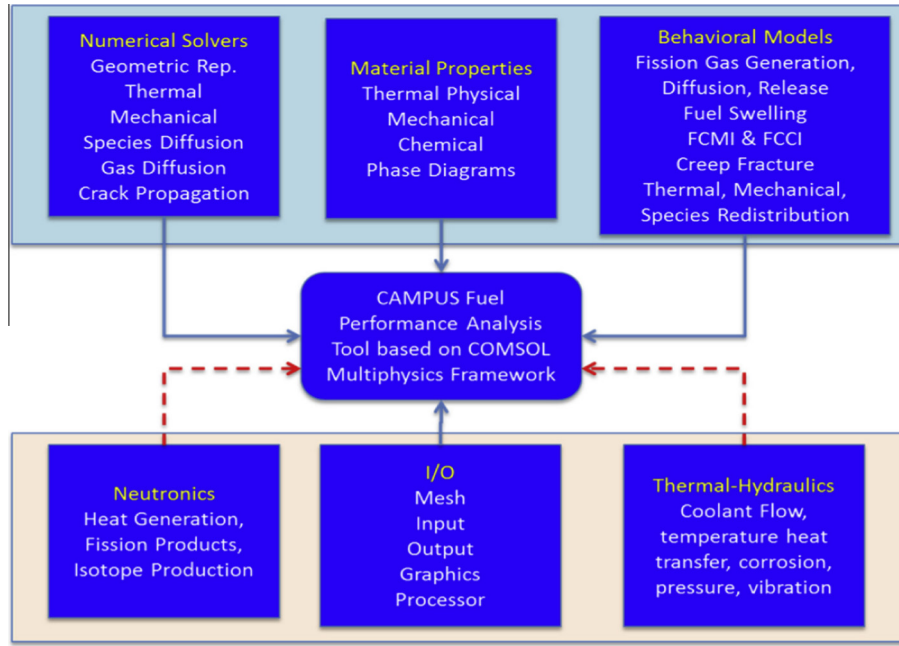


Fig. 1. CAMPUS fuel performance modeling and simulation structure.

Table 1
UO₂-SiC measured thermal conductivities (Ref. [8]).

SiC 5 vol%		SiC 10 vol%		SiC 20 vol%	
T (K)	k (W/m/K)	T (K)	k (W/m/K)	T (K)	k (W/m/K)
373.15	7.901	373.15	3.901	373.15	4.676
873.15	4.892	873.15	5.252	873.15	6.153
1173.15	3.306	1173.15	8.604	1173.15	10.297

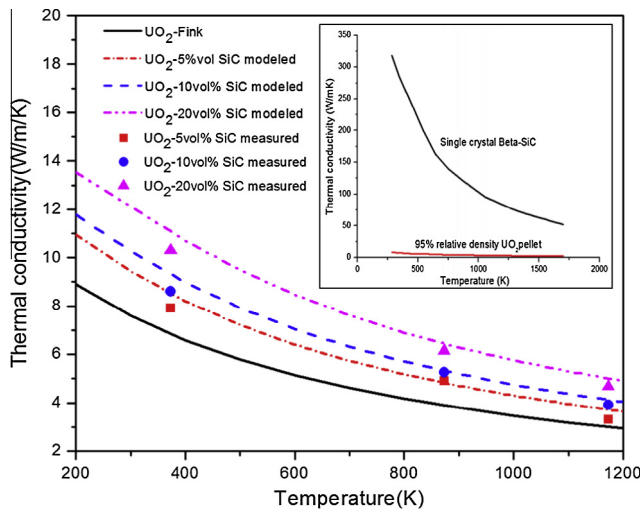


Fig. 2. Thermal conductivity of UO₂ and UO₂-SiC fuels from measured and modeled results, inset shows comparison of thermal conductivities between UO₂ and beta SiC from Ref. [8].

where v_l and v_t are the longitudinal and transverse phonon velocities. The referenced UO₂ longitudinal and transverse phonon velocities are 5552.7 and 2841.8 m/s [24], respectively, and those of SiC are 11,800 and 7600 m/s [25], respectively. Using these parameters with particle volume fraction (0–0.2), SiC particle radius (0.5 μ m) [26], and k_m from UO₂ and k_p from SiC into Hasselán and

Johnson model (Eq. (4)) the predicted thermal conductivity values are in excellent agreement compared to the measurements as illustrated in Table 1 and Fig. 2.

The correction factors in Eq. (1) are assumed to be same as for UO₂ fuel, and are defined as prescribed by Lucuta et al. [19], where:

$$f_d = \left(\frac{1.09}{bu^{3.265}} + 0.0643 \cdot \sqrt{\frac{T}{bu}} \right) \cdot \arctan \left(\frac{1.0}{\frac{1.09}{bu^{3.265}} + 0.0643 \cdot \sqrt{\frac{T}{bu}}} \right) \quad (6)$$

$$f_p = 1.0 + \left(\frac{0.019 \cdot bu}{3.0 - 0.019 \cdot bu} \right) \cdot \left(\frac{1.0}{1.0 + \exp \left(\frac{-(T-1200)}{100} \right)} \right) \quad (7)$$

$$f_{por} = \left(\frac{1.0 - p}{1.0 + 0.5 \cdot p} \right) \quad (8)$$

$$f_r = 1.0 - \frac{0.2}{1.0 + \exp \left(\frac{T-900}{80} \right)} \quad (9)$$

where T is the temperature in K, p is the porosity, and bu is the burn-up in at.%. The Fink-Lucuta model is valid from 298 to 3120 K [19,20] for pure UO₂ fuels. This provides a reasonable starting point for assessing the behavior of UO₂ composite fuel because of the large UO₂ volume fraction; however, in reactor and post irradiation measurements on composite fuel are required to assess the accuracy of this assumption and develop an improved models accounting for the presence of SiC.

2.1.2. Heat capacity

The heat capacity of composite material is a combination of weight fraction multiplied by single component heat capacity for both UO₂ and SiC. The relationship between the weight fraction and volumetric fraction is as following:

$$\begin{cases} W_{UO_2} = \frac{V_{UO_2} \cdot \rho_{UO_2}}{V_{UO_2} \cdot \rho_{UO_2} + V_{SiC} \cdot \rho_{SiC}} \\ W_{SiC} = \frac{V_{SiC} \cdot \rho_{SiC}}{V_{UO_2} \cdot \rho_{UO_2} + V_{SiC} \cdot \rho_{SiC}} \end{cases} \quad (10)$$

And then the $\text{UO}_2\text{-SiC}$ composite heat capacity is calculated as following:

$$C_p(\text{UO}_2 - \text{SiC}) = W_{\text{UO}_2} \cdot C_p(\text{UO}_2) + W_{\text{SiC}} \cdot C_p(\text{SiC}) \quad (11)$$

The MATPRO-11 correlation for the heat capacity of Urania was used in the simulation [27]:

$$C_p(\text{UO}_2) = \frac{K_1 \theta^2 \exp(\theta/T)}{T^2 [\exp(\frac{\theta}{T}) - 1]^2} + K_2 T + \frac{Y K_3 E_D}{2RT^2} \exp(-E_D/RT) \quad (12)$$

where T – temperature (K); K_1 – constant, 296.7 J/kg/K; K_2 – constant, 0.0243 J/kg K²; K_3 – constant, 8.75e7 J/kg; Y – oxygen to metal ratio; R = universal gas constant = 8.315 J/mol K; θ = Einstein temperature (K), for Urania = 535.285 K; E_D = activation energy for Frenkel defects (J/mol), for Urania = 1.577e5 J/mol. The correlation for the heat capacity of SiC from Xu et al. [28] was used in the simulation.

$$C_p(\text{SiC}) = \begin{cases} 925.65 + 0.3772 \cdot T - 7.9259 \times 10^{-5} \cdot T^2 - 3.1946 \times 10^7 \cdot T^{-2} & 200\text{K} \leq T \leq 2400\text{K} \\ 1365.54238 + 1.379 \times 10^{-3} \cdot T & 2400\text{K} \leq T \leq 2800\text{K} \end{cases} \quad (13)$$

The heat capacity of UO_2 , SiC, and $\text{UO}_2\text{-SiC}$ are shown in Fig. 3 assuming O/M ratio = 2.

2.1.3. Density

The density of $\text{UO}_2\text{-SiC}$ fuel is calculated using a combination of volumetric fraction multiplied by single component density for both UO_2 and SiC:

$$\rho(\text{UO}_2 - \text{SiC}) = V_{\text{UO}_2} \cdot \rho(\text{UO}_2) + V_{\text{SiC}} \cdot \rho(\text{SiC}) \quad (14)$$

The density of solid UO_2 recommended by Fink [20] was used in the simulation:

$$\rho(\text{UO}_2) = 10,970(a + bT + cT^2 + dT^3)^{-3}; \text{ with}$$

For $T \leq 923$ K:

$$a = 0.99734, \quad b = 9.802 \times 10^{-6}, \quad c = -2.705 \times 10^{-10}, \quad d = 4.391 \times 10^{-13} \quad (15)$$

For $T > 923$ K:

$$a = 0.99672, \quad b = 1.179 \times 10^{-5}, \quad c = -2.429 \times 10^{-9}, \quad d = 1.219 \times 10^{-12} \quad (16)$$

The density of solid SiC from Xu et al. [28] was used in the simulation:

$$\rho(\text{SiC}) = 3215 \times \left[1.0 - 3.0 \times \left(\frac{\Delta L}{L(298\text{K})} \right)_{\text{SiC}} \right] \quad 300\text{K} \leq T \leq 2800\text{K} \quad (17)$$

The densities of UO_2 , SiC and $\text{UO}_2\text{-SiC}$ are shown in Fig. 4.

2.1.4. Young's modulus, Poisson's ratio and thermal expansion coefficient

For this model we calculated the bulk properties of the $\text{UO}_2\text{-SiC}$ composite assuming a simple volume fraction weighting of the single component properties (Young's modulus, Poisson's ratio and thermal expansion coefficient) for both UO_2 and SiC, i.e.,

$$E(\text{UO}_2 - \text{SiC}) = V_{\text{UO}_2} \cdot E(\text{UO}_2) + V_{\text{SiC}} \cdot E(\text{SiC}) \quad (18)$$

$$\nu(\text{UO}_2 - \text{SiC}) = V_{\text{UO}_2} \cdot \nu(\text{UO}_2) + V_{\text{SiC}} \cdot \nu(\text{SiC}) \quad (19)$$

$$\frac{\Delta L}{L}(\text{UO}_2 - \text{SiC}) = V_{\text{UO}_2} \cdot \frac{\Delta L}{L}(\text{UO}_2) + V_{\text{SiC}} \cdot \frac{\Delta L}{L}(\text{SiC}) \quad (20)$$

Although this is a crude approximation for the elastic properties of composite materials, it is justified by the relative insensitivity of the model predictions to these parameters as a result of the relatively small elastic strains compared to the inelastic strains

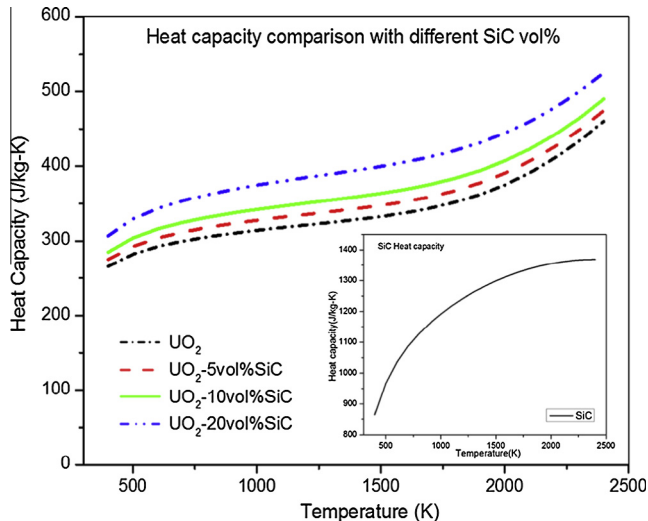


Fig. 3. Heat capacity of UO_2 , SiC and $\text{UO}_2\text{-SiC}$ with different SiC volume fraction assuming O/M ratio 2.

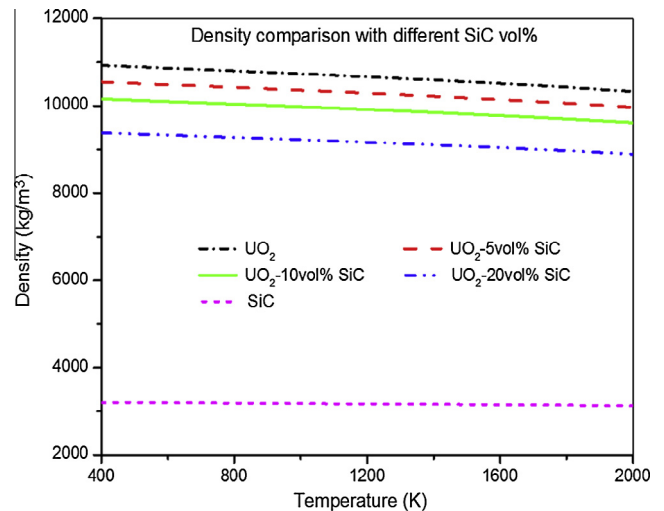


Fig. 4. Density of UO_2 , SiC and $\text{UO}_2\text{-SiC}$ with different SiC volume fraction.

(swelling, densification, thermal expansion and cracking). Young's modulus of solid UO_2 recommended by Martin [29] was used in the simulation:

$$E(\text{UO}_2) = ((2.334\text{e}11) \cdot (1 - (1.091\text{e} - 4) \cdot T \cdot \exp(-1.34X_{dev}))) \quad (21)$$

where X_{dev} – oxygen to metal ratio deviation. Poisson ratio of solid UO_2 recommended by Martin [29] was used in the simulation:

$$\nu(\text{UO}_2) = 0.316 + (0.5 - 0.316)(T - 300)/2800 \quad (22)$$

Young's modulus and Poisson ratio of solid SiC from Xu et al. [28] was used in the simulation:

$$E(\text{SiC}) = 428.3 - 0.04 \cdot T \cdot e^{-\frac{962}{T}} \text{ (GPa)} \quad (23)$$

$$\nu(\text{SiC}) = 0.160937533 - 2.84171929 \times 10^{-6} \cdot T \quad (24)$$

The thermal expansion of solid expansion of solid UO_2 recommended by Martin [29] was used in the simulation:

$$\frac{\Delta L}{L} = a + bT + cT^2 + dT^3 \quad (25)$$

For $T \leq 923$ K:

$$a = 9.828 \times 10^{-6}, \quad b = -6.390 \times 10^{-10}, \quad c = 1.33 \times 10^{-12}, \\ d = -1.757 \times 10^{-17}$$

For $T > 923$ K:

$$a = 1.183 \times 10^{-5}, \quad b = -5.103 \times 10^{-9}, \quad c = 3.756 \times 10^{-12}, \\ d = -6.125 \times 10^{-17}$$

The thermal expansion of solid SiC from Xu et al. [28] was used in the simulation:

(5) Young's modulus

$$E_{\text{Zr}} = 108.8 \times 10^9 - 5.475 \times 10^7 T \quad (32)$$

(6) Thermal expansion

$$\alpha_{\text{Zr}} = \begin{cases} 7.092 \times 10^{-6}, & \text{radial} \\ 9.999 \times 10^{-6}, & \text{angular} \\ 5.458 \times 10^{-6}, & \text{axial} \end{cases} \quad (33)$$

2.3. Gap properties

The gap between fuel and cladding is filled with helium gas, with the properties obtained from Heat Transfer by Bejan [32].

(1) Helium density

$$\rho_{\text{He}} = 0.0818 - 8.275 \times 10^{-15}(T - 600) \text{ (J/kg K)} \quad (34)$$

(2) Helium heat capacity

$$C_{p,\text{He}} = 5190 \text{ (J/kg K)} \quad (35)$$

(3) Helium thermal conductivity

$$k_{\text{He}} = 0.04679 + 3.81 \times 10^{-4}T - 6.786 \times 10^{-8}T^2 \text{ (W/m K)} \quad (36)$$

3. Model implementation

3.1. Model geometry

The model geometry in this work adopted a 2D axisymmetric plane with a Zr-4 cladding (see Fig. 5(a)). A single pellet was

$$\left(\frac{\Delta L}{L(298 \text{ K})} \right)_{\text{SiC}} = \begin{cases} -1.1753 \times 10^{-4} - 1.8276 \times 10^{-6} \cdot T + 8.9 \times 10^{-9} \cdot T^2 - \\ 5.18133 \times 10^{-12} \cdot T^3 + 1.13115 \times 10^{-15} \cdot T^4 & 300 \text{ K} \leq T \leq 1673 \text{ K} \\ 4.26118 \times 10^{-3} + (4.97628 \times 10^{-6} \cdot T - 6.33555 \times 10^{-3}) & 1673 \text{ K} \leq T \leq 2800 \text{ K} \end{cases} \quad (26)$$

2.1.5. Oxygen diffusion coefficient is defined as below in the simulation [30]

$$D_0 = 1.39\text{e} - 6 \cdot \exp(-9128/T) \quad (27)$$

2.2. Zircaloy cladding properties

Zircaloy cladding properties were adopted from IAEA thermo-physical properties database report [31].

(1) Cladding thermal conductivity

$$k_{\text{Zr}} = 11.498 + 4.6765 \times 10^{-3} \cdot T + 2.761 \times 10^{-6} \cdot T^2 \\ + 2.2147 \times 10^{-6} \cdot T^3 \text{ (W/m K)} \quad (28)$$

(2) Heat capacity

$$C_{p,\text{Zr}} = (255.66 + 0.1024T) \text{ (J/kg K)} \quad (29)$$

(3) Density

$$\rho_{\text{Zr}} = 6595.2 - 0.1477T \text{ (kg/m}^3\text{)} \quad (30)$$

(4) Poisson ratio

$$\nu_{\text{Zr}} = 0.32 \quad (31)$$

considered to represent all the pellets with a periodic boundary condition in the axial direction, as shown in Fig. 5(b), with a mapped mesh.

3.2. Heat generation and conduction

The heat transfer model was implemented using the “Heat transfer in Solids” physics module included in COMSOL. This module defines the temperature, T (K), by the heat conduction equation:

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q \quad (37)$$

where ρ , C_p and k are the material properties of density, specific heat capacity and thermal conductivity, respectively. We can define all the material properties in the “Materials” module. The volumetric heating term, Q , accounts for heat produced in the fuel. For the single pellet case, we considered a uniformly distributed heat source term, which is given by:

$$Q = \frac{P_{\text{lin}}}{\pi \cdot a_{\text{pellet}}^2} \quad (38)$$

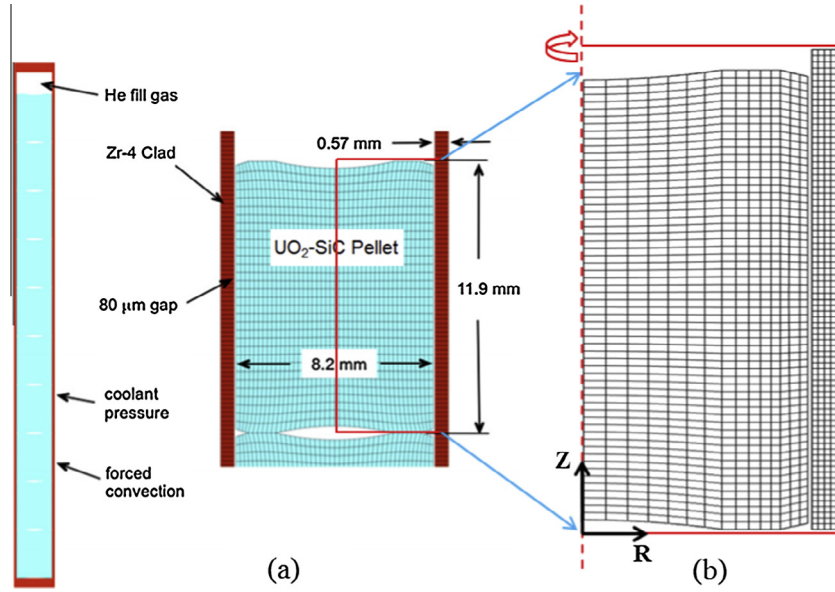


Fig. 5. (a) Geometry, material and typical mesh used in BISON, and (b) 2D-axisymmetric one pellet geometry and mesh used in this work.

where P_{lin} is the linear power, a_{pellet} is fuel pellet radius. This neglects the effect of neutron flux depression which causes a greater portion of the power to be produced closer to the outside surface of the fuel and therefore slightly over estimates the temperatures (on the order of 10–50 K).

The calculation of fuel burnup is given as:

$$\frac{dBu}{dt} = \frac{Q}{3.6 \times 10^9 \rho_U} = \frac{Q}{3.6 \times 10^9 \left(\frac{238 \rho_{UO_2}}{238 + 16(2 + X_{dev})} \right)} \quad (39)$$

Q is the volumetric heat source term from Eq. (38) in unit of W/m³, ρ_{UO_2} is the density of the fuel in unit of kg/m³, X_{dev} is deviation from stoichiometry of oxygen to metal ratio and Bu is in unit of MW h/kg-U.

Heat transport across the pellet-to-cladding gap is modeled assuming one-dimensional steady-state heat transfer by using the following radial heat flux:

$$Q_r = (h_{gas} + h_{solid} + h_{rad})(T_{fuel} - T_{cladding}) \quad (40)$$

where the total conductance across the gap is the sum of gas conductance (h_{gas}), conductance due to solid-solid contact (h_{solid}) and the conductance due to radiant heat transfer (h_{rad}). This model is applied between the fuel and clad, following the modeling approach used in BISON and FAST.

The gas conductance h_{gas} is described using the form proposed by Ross and Stoute [33]:

$$h_{gas} = \frac{k_{gas}(T_{gas})}{d_g + C_r(r_1 + r_2) + g_1 + g_2} \quad (41)$$

where k_{gas} is the conductivity of the gas in the gap, d_g is the gap width (computed dynamically according to deformation predicted via solid-mechanics), C_r is a roughness coefficient with r_1 and r_2 the roughnesses of the two surfaces, and g_1 and g_2 are jump distances at the two surfaces.

The conductance due to solid-solid contact, h_{solid} , is given by using the empirical model suggested by Olander [34]:

$$h_{solid} = C_s \frac{2k_1 k_2}{k_1 + k_2} \frac{P_c}{\delta^{0.5} H} \quad (42)$$

where C_s is a constant (typically 1.0), k_1 and k_2 are the thermal conductivities of the solid materials in contact, P_c is the contact pressure which is modeled by using a penalty method, $\delta^{0.5}$ is the

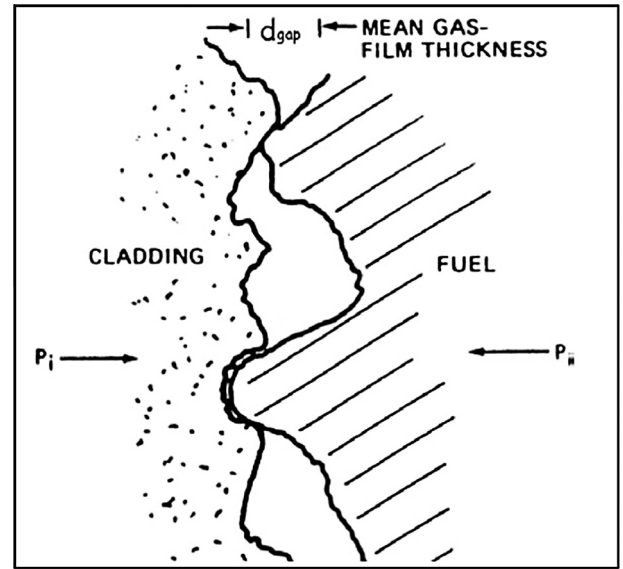


Fig. 6. Closed fuel-cladding gap from Ref. [34].

average radius of solid-solid contact, δ is approximated as $\delta = 0.8(r_1 + r_2)$, a function of roughness of fuel outer surface and cladding inner surface, and H is the Meyer hardness of the softer material.

The conductance due to radiative heat transfer, h_{rad} , is calculated using a 1D infinite-parallel plates approximation (i.e., assuming the view factor equals 1). Based on the Stefan-Boltzmann law:

$$q_{rad} = \sigma \frac{1}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} (T_1^4 - T_2^4) \equiv h_{rad}(T_1 - T_2) \quad (43)$$

where σ is the Stephan-Boltzman constant, ϵ_1 and ϵ_2 are the emissivities of the radiating surfaces, T_1 and T_2 are the temperatures of the radiating surfaces. The radiative conductance can be written in the form:

$$h_{rad} = \sigma \frac{1}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} (T_1^2 + T_2^2)(T_1 + T_2) \quad (44)$$

As discussed by Olander [34], two types of gaps were proposed for the conductance of the fuel-cladding gap: open gap and closed gap. The open gap is defined as the state that the fuel and cladding are not in physical contact, and the mechanism of heat transfer is by conduction through the filling gas and radiation through electromagnetic wave. As we can see from Eqs. (40)–(44), the h_{solid} term can be ignored as the contact pressure P_c is zero under an open gap condition.

When the fuel-cladding gap is closed as a result of thermal expansion, swelling and creep of the fuel and the cladding, this situation is called the closed gap. The heat is transported by solid conduction through the regions on the fuel and cladding surfaces that are in physical contact as well as by conduction through the gas film that fills the portion of the interface where contact is not made. Fig. 6 suggests what closed gap might look like at sufficient magnification. In the closed gap, the contact area is still a relatively small fraction compared to the area filled with gas. So, the gas conductance h_{gas} is considered to apply to the whole heat transfer area for both open and closed cases. Similarly, an effective (averaged) value is considered for the solid-solid contact conductance, h_{solid} , which distributes the local solid-solid contact conductance resulting from the surface roughness over the macroscopic contact area. Ross and Stoute [33] found that a value for the constant C_s of approximately unity gave the best fit to their measurements. And the radiation conductance can be ignored under the closed gap situation as the fuel outer surface temperature (T_1) and cladding inner surface temperature (T_2) are almost equal. This way we achieve a reasonable solution to the complex thermal contact heat transfer, and the model can fit the experimental results [33] well at the same time.

The heat conductance from the fuel cladding to the bulk coolant is considered as thin-film heat transfer, the net radial heat flux from the cladding to the bulk coolant, Q_r , is calculated as:

$$Q_r = \frac{T_{cladding} - T_{coolant}}{((t_{oxide}/k_{oxide}) + (1/h_{film}))} - Q_{oxide} \quad (45)$$

where $T_{cladding}$ is cladding temperature and $T_{coolant}$ is coolant temperature; k_{oxide} is the thermal conductivity of zirconium-oxide layer and t_{oxide} is the layer thickness; h_{film} is the effective convective heat transfer coefficient from the oxide surface to the bulk coolant (i.e., through the fluid surface film) calculated from thermal-hydraulic codes accounting for the coolant conditions; and Q_{oxide} is heat flux resulting from the cladding oxidation and is given by Leistikov and Schanz [35]:

$$Q_{oxide} = \frac{42.426 \times 10^9}{1.56} \frac{dt_{oxide}}{dt} \quad (46)$$

where zirconium-oxide layer thickness t_{oxide} is based on the corrosion rate. The MATPRO-10 correlation for cladding oxidation was implemented on the outside surface of the fuel cladding. For the pre-transition period, the corrosion rate is given by an Arrhenius equation [35]:

$$\frac{dt_{oxide}^3}{dt} = C_1 \exp\left(\frac{-Q_1}{RT_i}\right), \quad \text{for } t_{oxide} \leq t_{trans} \quad (47)$$

For the post-transition period, the corrosion rate is given by an Arrhenius equation [36]:

$$\frac{dt_{oxide}}{dt} = C_2 \exp\left(\frac{-Q_2}{RT_i}\right), \quad \text{for } t_{oxide} > t_{trans} \quad (48)$$

where C_1 , C_2 and Q_1 , Q_2 are the rate constants and activation energy for pre-transition oxidation and post-transition oxidation, respectively.

3.3. Oxygen diffusion model

A hyperstoichiometric model for oxygen diffusion is adopted from Newman et al. [37], which is also used in the BISON code. In this case J denotes the oxygen flux within the hyperstoichiometric regime with,

$$J = -D \left(\nabla C + \frac{CQ^*}{FRT^2} \nabla T \right) \quad (49)$$

where D is diffusivity of UO_2 , F is the thermodynamic factor of oxygen, C is the atomic fraction of oxygen interstitials and R is the universal gas constant. The oxygen flux J satisfies the equation of continuity expressing the oxygen atoms conservation as:

$$\frac{\partial C}{\partial t} + \nabla \cdot J = 0 \quad (50)$$

3.4. Deformation mechanics

3.4.1. Elastic deformation

In the reactor, the fuel element deformation behavior is described by using Cauchy's equation in the BISON code, which we can also find in the COMSOL structural mechanics module given as the following form:

$$-\nabla \cdot \sigma = F_v \quad (51)$$

where σ is the Cauchy stress tensor and F_v is the body force per unit volume, which depends on applied forces, thermal expansion, materials creep, fuel densification and fission gas swelling. The stress is calculated with a liner elastic constitutive model:

$$\sigma = [C][\varepsilon] \quad (52)$$

where C is the material matrix, ε is the elastic strain vector. The elastic strain is calculated as the total strain, defined as $1/2[\nabla \cdot \mathbf{u} + \nabla \cdot \mathbf{u}^T]$, ($\mathbf{u}$ is the displacement vector) minus any sources of inelastic strain. In the 2D axisymmetric case, as used in this paper, the strain components are given by:

$$\begin{aligned} \varepsilon_r &= \frac{\partial u}{\partial r}, \quad \varepsilon_\theta = \frac{u}{R}, \quad \varepsilon_z = \frac{\partial v}{\partial z}, \quad \varepsilon_{rz} = \frac{1}{2} \left(\frac{\partial v}{\partial r} + \frac{\partial u}{\partial z} \right), \quad \varepsilon_{r\theta} \\ &= \varepsilon_{\theta z} = 0 \end{aligned} \quad (53)$$

We adopted the penalty method from the FAST code to model the pellet-to-cladding contact with a force applied normal to the cladding. The penalty function was selected such that the interface pressure, P_i , increases exponentially to the estimated contact pressure (P_{est}) as the fuel and cladding boundaries come into contact and then increases linearly with the penetration distance (d_{pen}) at a rate determined by the penalty factor, P_f . This penalty method formulation prevents the extremely high stiffness values which could occur if the exponential term was still used once penetration occurs [17].

$$P_i = \begin{cases} P_{est} \exp\left(\frac{P_f d_{pen}}{P_{est}}\right) & d_{pen} < 0 \\ P_{est} + P_f d_{pen} & d_{pen} \geq 0 \end{cases} \quad (54)$$

Both the estimated contact pressure and the penalty factor are adjustable parameters. For results of the light water reactor case presented in this work, values of $P_{est} = 1$ MPa and $P_f = 5 \times 10^{12}$ Pa/m were used for the pellet-to-cladding contact. This penalty contact model used the COMSOL built-in contact module and we set the contact tolerance as 10 μm , which is close to the fuel and cladding surface roughness. In this work, the pellet-to-pellet contact and the friction between the pellet-to-cladding surfaces are neglected. Modeling these phenomena would require a very signif-

ificant increase of the computation complexity of the code and while producing a relatively small impact on the solution (relative to other neglected phenomena such as pellet cracking).

3.4.2. Thermal strain/expansion

The thermal strain model in Eq. (20) was used in this work considering an isotropic UO₂-SiC fuel pellet.

3.4.3. Fuel densification and fission product swelling

Fuel densification is computed by using the ESCORE empirical model given by Rashid et al. [38]:

$$\varepsilon_D = \Delta\rho_0 \left(e^{\frac{Bu_D(0.01)}{C_D^{Bu_D}}} - 1 \right) \quad (55)$$

where ε_D is the densification strain, $\Delta\rho_0$ is the total densification which can occur, and Bu_D is the burnup when densification is completed. For temperatures below 750 °C the parameter C_D is given by 7.2–0.0086 (T–25); above 750 °C it is 1.0 (T in °C). Swelling due to both solid and gaseous fission products is considered by using empirical relations from MATPRO [39]. The volumetric strain due to solid fission product swelling is given as a simple linear function of burnup:

$$\frac{\delta\varepsilon_{sw-s}}{\delta t} = 5.577 \times 10^{-5} \rho \frac{\delta Bu}{\delta t} \quad (56)$$

where ε_{sw-s} is the volumetric solid swelling, Bu the local burnup (fissions/atoms-U), and ρ is the density of uranium (kg U/m³). While swelling due to gaseous fission products is calculated by a semi-empirical model from MATPRO [39]:

$$\frac{\delta\varepsilon_{sw-g}}{\delta t} = 1.96 \times 10^{-31} \rho \frac{\delta Bu}{\delta t} (2800 - T)^{11.73} e^{-0.0162/(2800-T)} e^{-0.0178\rho Bu} \quad (57)$$

where ε_{sw-g} is the volumetric gas swelling.

3.4.4. Fuel thermal and radiation creep

Since no creep rate data is available for the UO₂-SiC composite fuel, a model for single component UO₂ has been substituted since it is the dominate phase. The MATPRO FCREEP model [39] for combined secondary thermal creep and irradiation creep of UO₂ is used in this work, given as:

$$\dot{\varepsilon} = \frac{A_1 + A_2 \dot{F}}{(A_3 + D_f) G^2} \sigma e^{\left(\frac{-Q_1}{RT}\right)} + \frac{A_4}{(A_6 + D_f)} \sigma^{4.5} e^{\left(\frac{-Q_2}{RT}\right)} + A_7 \dot{F} \sigma e^{\left(\frac{-Q_3}{RT}\right)} \quad (58)$$

where $\dot{\varepsilon}$ is the creep rate (1/s), σ is the effective (Mises) stress (Pa), T is the temperature (K), D_f is the fuel density (in percent of theoretical), G is the grain size (μm), $\dot{F}$ is the volumetric fission rate (fissions/m³/s), Q_i are the activation energies (J/mol), R is the universal gas constant and A_1 to A_7 are material constants, detailed parameters can be found in the BISON code [15].

3.4.5. Cladding thermal and radiation creep

We considered Zircaloy as the cladding, and the secondary thermal creep model recommended by Hayes and Kassner [40] as shown below:

$$\dot{\varepsilon}_{ss} = A_0 \left(\frac{\sigma_m}{G} \right)^n \exp \left(\frac{-Q}{RT} \right) \quad (59)$$

where $\dot{\varepsilon}_{ss}$ is the effective thermal creep rate (1/s), σ_m is the effective (Mises) stress (Pa), Q is the activation energy (J/mol), R is the universal gas constant (J/mol/K), T is the temperature (K), G is the shear modulus (Pa) recommended by Moon et al. [41] as $G = 4.2519 \times 10^{10} - 2.2185 \times 10^7 T$, n is specified by Hayes and Kassner as a power of 5, A_0 is the same as that used in BISON with

a value of 3.14×10^{24} (1/s). Irradiation-induced cladding creep is implemented from Hooke [42], which relates the creep rate to the current fast neutron flux and stress:

$$\dot{\varepsilon}_{ir} = C_0 \Phi^{C_1} \sigma_m^{C_2} \quad (60)$$

where $\dot{\varepsilon}_{ir}$ is the effective irradiation creep rate (1/s), Φ is the fast neutron flux (N/m² s), σ_m is the effective stress (MPa), and C_0 , C_1 and C_2 are material constants. We also assume an effective (Mises) stress (same as BISON) rather than the original Hoppe's circumferential stress formulation.

3.5. Fission gas production and release

The fission gas release (FGR) is calculated by using the Forsberg-Massih two-stage model [43], namely, in the first stage, the fission gas is produced in the fuel grains and diffuses to the fuel grain boundary and then forms intergranular lenticular bubbles. The second stage occurs when the fuel grain surface gets saturated and any additional fission gas at the grain surface is released to the free volume of the element. These models have been developed for traditional UO₂ fuel with a single phase (no SiC). Assuming the two phases remain well separated, it is reasonable that the behavior within the UO₂ grains would remain roughly unchanged. Similarly, the grain boundary behavior between UO₂ two grains would likely remain. However, the behavior of the fission gas at the UO₂-SiC interfaces remains unknown and is not accounted for in this model.

3.5.1. Fission gas release to the grain boundaries

We adopted the numerical solution method of the FAST code to implement a Booth type diffusion model [44], which is described in the following equation form:

$$\frac{\partial C}{\partial t} = -D \nabla^2 C + Q_{fg} \quad (61)$$

where C is the concentration of fission gas atoms in the fuel grains, D is the diffusion coefficient of fission gas in the fuel grain and Q_{fg} is the volumetric rate of production of fission gas atoms. In the FAST code, a two-dimensional rectangular domain is used in which one direction represents the radial coordinate of the grain within the fuel pellet and the other direction represents the radial coordinate within the spherical grain. Eq. (26) was transformed to a non-dimensional form by substitution of $\eta = \frac{r}{g_r}$ to obtain:

$$\eta^2 g_r^2 \frac{\partial C}{\partial t} = \frac{\partial C}{\partial \eta} \left(-D \eta^2 \frac{\partial C}{\partial \eta} \right) + \eta^2 g_r^2 Q_{fg} \quad (62)$$

Then the release rate to the grain boundary, R_{gb} , is

$$R_{gb} = \frac{3}{g_r^2} D \frac{\partial C}{\partial \eta} \Big|_{\eta=1} \quad (63)$$

where g_r is the local fuel grain radius, the local UO₂ grain size was determined by a simple empirical model [45] as implemented in BISON. The kinetics of grain growth is described by the equation:

$$\frac{dg_d}{dt} = k \left(\frac{1}{g_d} - \frac{1}{g_{max}} \right) \quad (64)$$

where g_d is the average grain diameter ($2g_r$), t (s) the time, k (m²/s) the rate constant, which is given by:

$$k = 1.46 \times 10^{-8} \exp \left(\frac{-32,100}{T} \right) \quad (65)$$

T is the temperature in K, g_{max} is the limiting grain size (m) as a function of temperature, which is:

$$g_{max} = 2.23 \times 10^{-3} \exp \left(\frac{-7620}{T} \right) \quad (66)$$

3.5.2. Fission gas release to the fuel element

After the fission gas has been released to the grain surface, it will be trapped in intergranular bubbles among fuel grains. Once the gases accumulated on grain boundaries reaches the saturated gas concentration, G_{bsat} , the fission gas will release to the fuel free volume. The grain boundary saturation, G_{bsat} (atoms/m³), is given by White and Tucker [44] as:

$$G_{bsat} = \frac{4r_f f(\theta_{fg}) f_B}{3k_B T \sin^2(\theta_{fg})} \left(P_{ext} + \frac{2\gamma_{se}}{r_f} \right) \left(\frac{3}{g_r} \right) \quad (67)$$

$$f(\theta) = 1 - 1.5 \cos(\theta) + 0.5 \cos^3(\theta) \quad (68)$$

Here, $r_f = 5 \times 10^{-7}$ m is the grain boundary bubble radius, $f(\theta_{fg})$ is a function that accounts for the shape of the bubbles, $\theta_{fg} = 50^\circ$ is the semi-dihedral angle between bubble surfaces, $f_B = 0.5$ is the grain boundary fraction at saturation, k_B is the Boltzmann constant, T is the temperature in K, $\gamma_{se} = 0.626$ J m⁻² is the surface tension of the bubbles, and g_r is the grain radius. All these parameters are the same as those used in BISON. An additional approximation $P_{ext} = 0$ is the externally applied hydrostatic pressure, which is assumed to be zero for simplicity (since accurately calculating the hydrostatic pressure requires accounting for the cracking behavior of the UO₂ which is not included in this model). This is an area for future improvement.

3.6. Gap/plenum pressure

The pressure in the gap and plenum is computed based on the ideal gas law:

$$P = \frac{nR}{\int_V \frac{1}{T} dV} \quad (69)$$

where P is the gap/plenum pressure, n is the moles of gas, R is the ideal gas constant, T is the temperature and V is the volume of the cavity. As in FAST, the gas volume is divided into sub-volumes, which are calculated individually based on the temperatures at the boundaries of the sub-volume and added together. The temperature T is calculated as the average temperature of the pellet exterior and cladding interior surfaces.

4. Results and discussions

In this section, the modeling results of enhanced thermal conductivity UO₂-SiC fuel behavior in CAMPUS are presented for a 2D axisymmetric LWR fuel rodlet. The CAMPUS code predictions are compared with those from BISON, ABAQUS and FRAPCON (see Fig. 7) to verify and validate the CAMPUS code capability. The CAMPUS code is then subsequently applied to predict the behavior of enhanced thermal conductivity UO₂-SiC fuels with

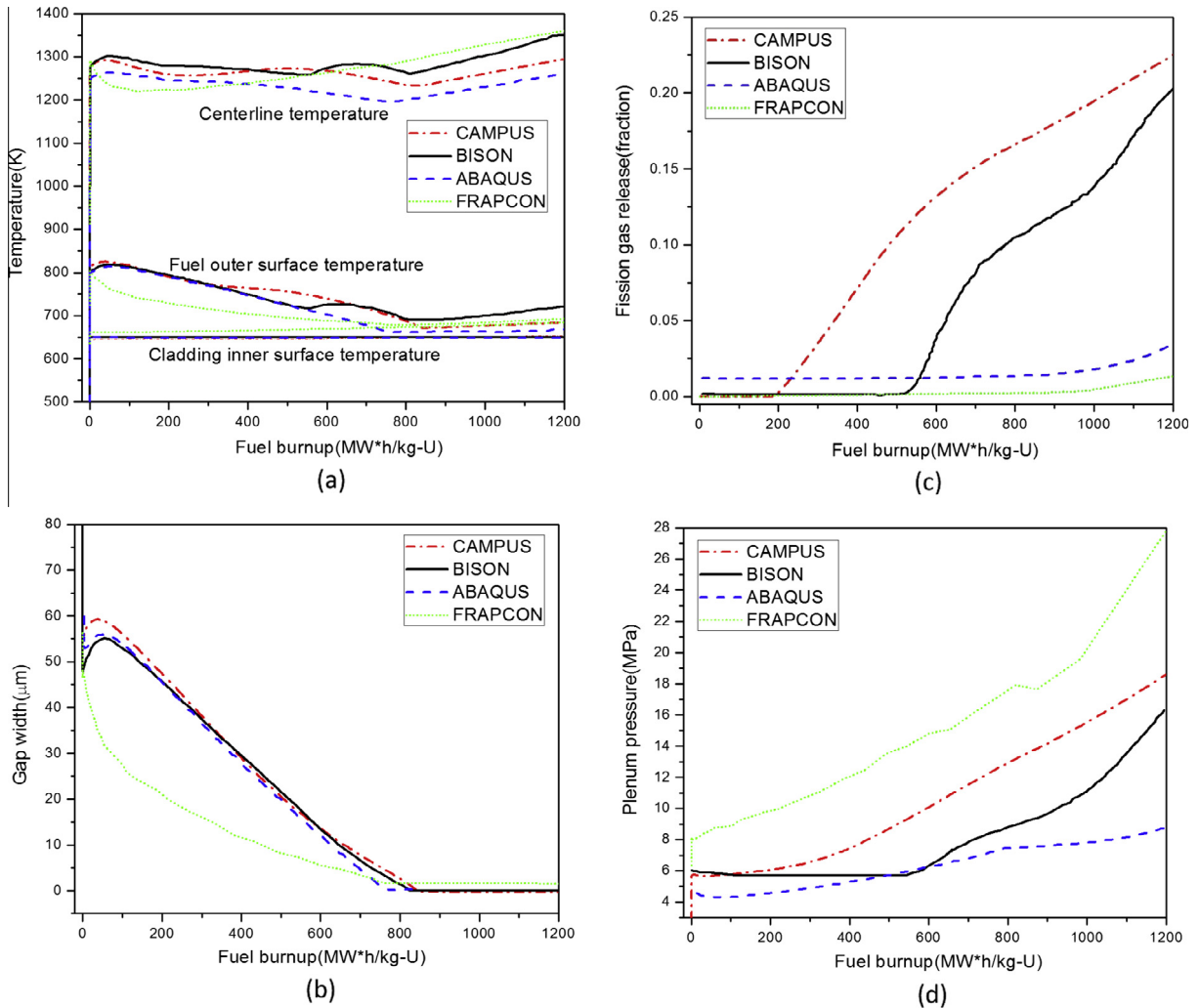


Fig. 7. Comparison of CAMPUS code to BISON, ABAQUS and FRAPCON. (a) Temperature profile; (b) gap width; (c) fission gas release; and (d) plenum pressure.

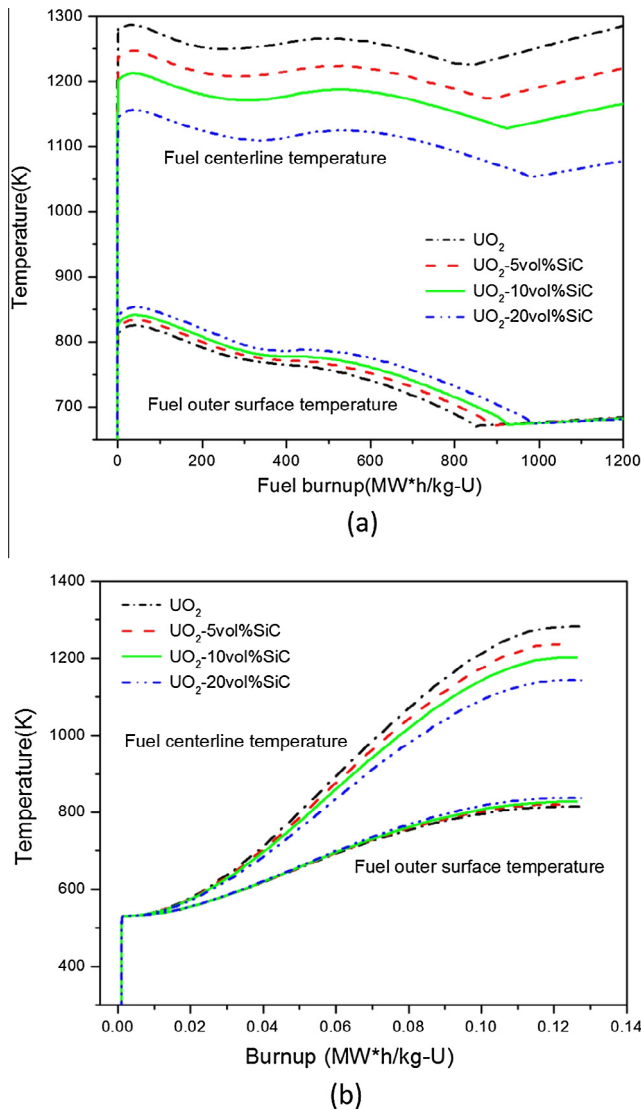


Fig. 8. (a) Calculated temperature evolutions at fuel centerline, outer surface with: (1) UO₂ fuel and Zircaloy cladding; (2) UO₂-5 vol%SiC fuel and Zircaloy cladding; (3) UO₂-10 vol%SiC fuel and Zircaloy cladding; and (4) UO₂-20 vol%SiC fuel and Zircaloy cladding. (b) Calculated temperature evolutions within first three hours at fuel centerline, outer surface with: (1) UO₂ fuel and Zircaloy cladding; (2) UO₂-5 vol%SiC fuel and Zircaloy cladding; (3) UO₂-10 vol%SiC fuel and Zircaloy cladding; and (4) UO₂-20 vol%SiC fuel and Zircaloy cladding.

different volume fractions of SiC, and the results obtained from these cases are discussed in the following section (see Figs. 8–15).

For all of the modeling simulations presented here, a 2D axisymmetric simplified fuel pellet was modeled with a typical finite element mesh as shown in Fig. 5(b). The geometry includes an individual pellet (UO₂-SiC or UO₂), Zr-4 cladding, and initial 80 μ m pellet-cladding gap, and computationally counted upper plenum. The plenum to fuel length ratio was set to 0.045. A uniform convective boundary at the cladding outer wall is used to simulate the heat transfer from the cladding to the outer flowing coolant. The linear power is assumed to rise linearly over three hours and then is held constant for 3.8 years. Typical PWR operating conditions were used as given in Table 2.

To verify and validate the CAMPUS code simulation capability, a comparison of the CAMPUS code predictions was performed for UO₂ fuel behavior against those from BISON, ABAQUS and FRAPCON codes. Overall, the CAMPUS code shows reasonable agreement with the other fuel codes, with discrepancies likely due to

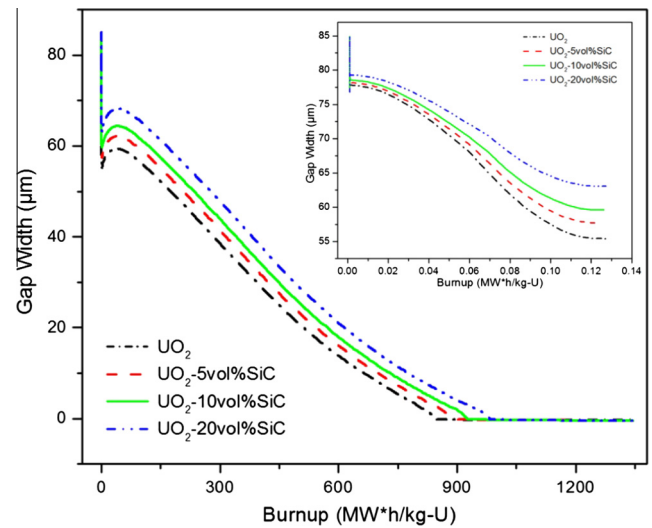


Fig. 9. Calculated gap size evolution with different type of fuels and Zircaloy cladding, inset shows the gap size evolutions of the first three hours.

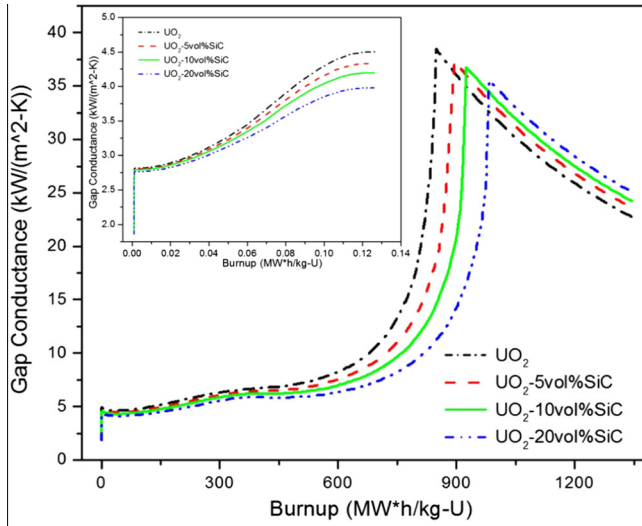
differences in models. Fig. 7(a) shows the predicted temperatures at the fuel centerline, fuel outer surface and cladding inner surface during power up and steady state operation with four codes. The CAMPUS predicted temperature profiles are found to be consistent with those from the other codes. Some differences among the code predictions attribute to the models employed, for example, fuel densification and fission gas release models. As shown in Fig. 7 (b), the CAMPUS gap size evolution is very similar to that from BISON and ABAQUS, while the FRAPCON predicted gap size decrease dramatically and results in an earlier gap closure due to a different mechanical model (fuel densification) used.

As shown in Fig. 7(c), the fission gas release from CAMPUS was earlier than from BISON, and the release fraction was a little bit larger than BISON, which further induced a higher plenum pressure as shown in Fig. 7(d), pushing the cladding outward and a delayed gap closure, owing to different fission gas release models used in BISON and CAMPUS. The fission gas release and plenum pressure, however, follow similar evolution trends by all the four codes.

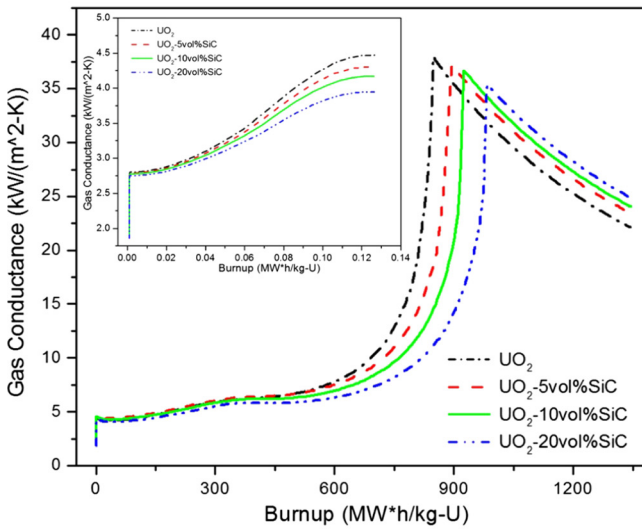
The CAMPUS code was then subsequently applied to predict the behavior of enhanced thermal conductivity UO₂-SiC fuels with different volume fractions of SiC, Fig. 8 shows the predicted temperature evolutions by CAMPUS code at the fuel centerline, fuel outer surface and cladding inner surface during first three hours power up (Fig. 8(a)) and steady operation (Fig. 8(b)), respectively. The fuel temperature is found to increase rapidly during the initial power-up as thermal equilibrium is established. This is followed by a gradual increase due to densification increasing pellet-cladding gap distance reducing heat conduction. Eventually fuel swelling dominates the densification leading to gap closure and a gradual decrease in fuel temperatures. Finally, the fuel temperature begins to increase again as the thermal conductivity degrades with increasing burnup.

From Fig. 8, we can see that the fuel temperature can be significantly lowered using the enhanced thermal conductivity UO₂-SiC fuel. For the case of UO₂-20 vol%SiC fuel and Zircaloy cladding in Fig. 8, the fuel centerline temperature was lowered about 140 K compared to the UO₂ fuel. However, there is no significant difference for UO₂-SiC fuel outer surface temperature compared to UO₂ fuel and Zircaloy cladding as shown in Fig. 7.

The gap size evolution is shown in Fig. 9 for the four different cases: UO₂ fuel, UO₂-5 vol%SiC fuel, UO₂-10 vol%SiC fuel, and UO₂-20 vol%SiC fuel. The gap size is a combined effect of fuel swelling, densification, thermal expansion and cladding creep down.



(a)



(b)

Fig. 10. Calculated evolution of total gap conductance (a) and gas conductance (b) with different type of fuels and Zircaloy cladding, inset shows the corresponding gap conductance evolutions of the first three hours.

We can see the initial gap size was about 80 μm , at the start of irradiation the gap size decreases sharply due to thermal expansion of the pellet as it reaches operational temperature (as we can see from the inset of Fig. 9). The gap size is then increased for a short period due to densification of the fuel, although this phenomenon eventually saturates and fuel swelling causes the gap to decrease until the fuel and cladding are in contact. From Fig. 9, the closure of the gap is then observed in order of increasing SiC content: UO_2 , UO_2 -5 vol%SiC, UO_2 -10 vol%SiC and UO_2 -20 vol%SiC. The SiC addition not only increases the fuel thermal conductivity, which lowers the fuel temperature, but also lowers the volume fraction of the fuel to swell (which is significant for the fission gas swelling is non-linear with burnup).

The total gap conductance and gas conductance are shown in Fig. 10(a) and (b), respectively. And no significant difference was found between the total gap conductance and gas conductance. The total conductance across the gap is the sum of gas conductance (h_{gas}), conductance due to solid-solid contact (h_{solid}) and conductance due to radiant heat transfer (h_{rad}), which is a combined effect

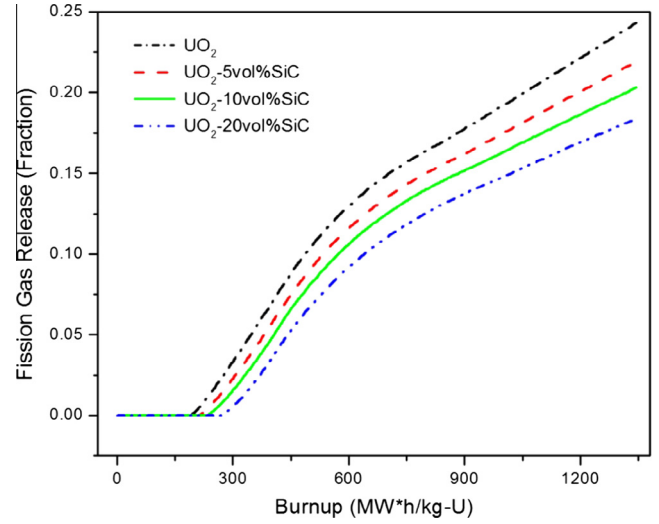


Fig. 11. Calculated fission gas release evolution with different type of fuels and Zircaloy cladding.

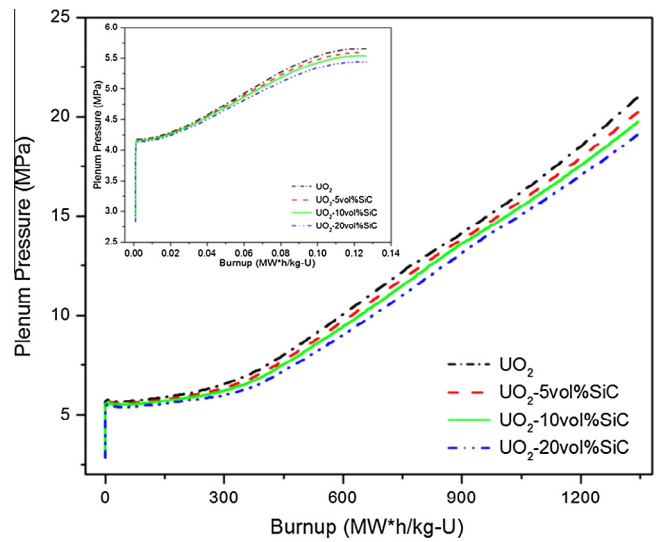


Fig. 12. Calculated plenum pressure evolution with different type of fuels and Zircaloy cladding, inset shows the plenum pressure evolutions of the first three hours.

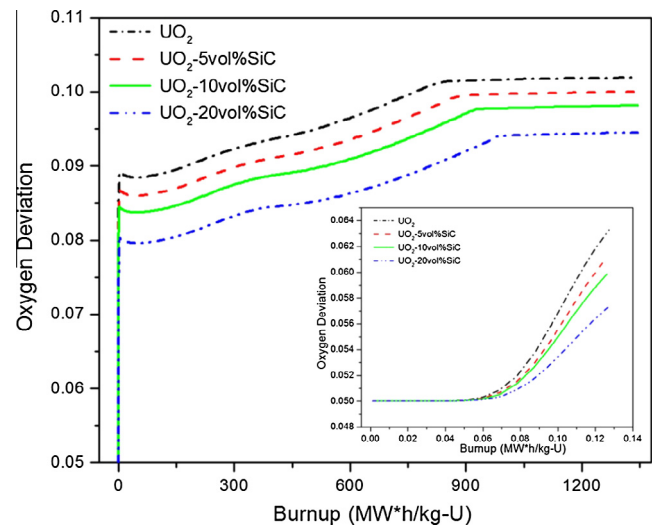


Fig. 13. Calculated oxygen deviation evolution of fuel centerline with different type of fuels and Zircaloy cladding, inset shows the oxygen deviation evolutions of the first three hours.

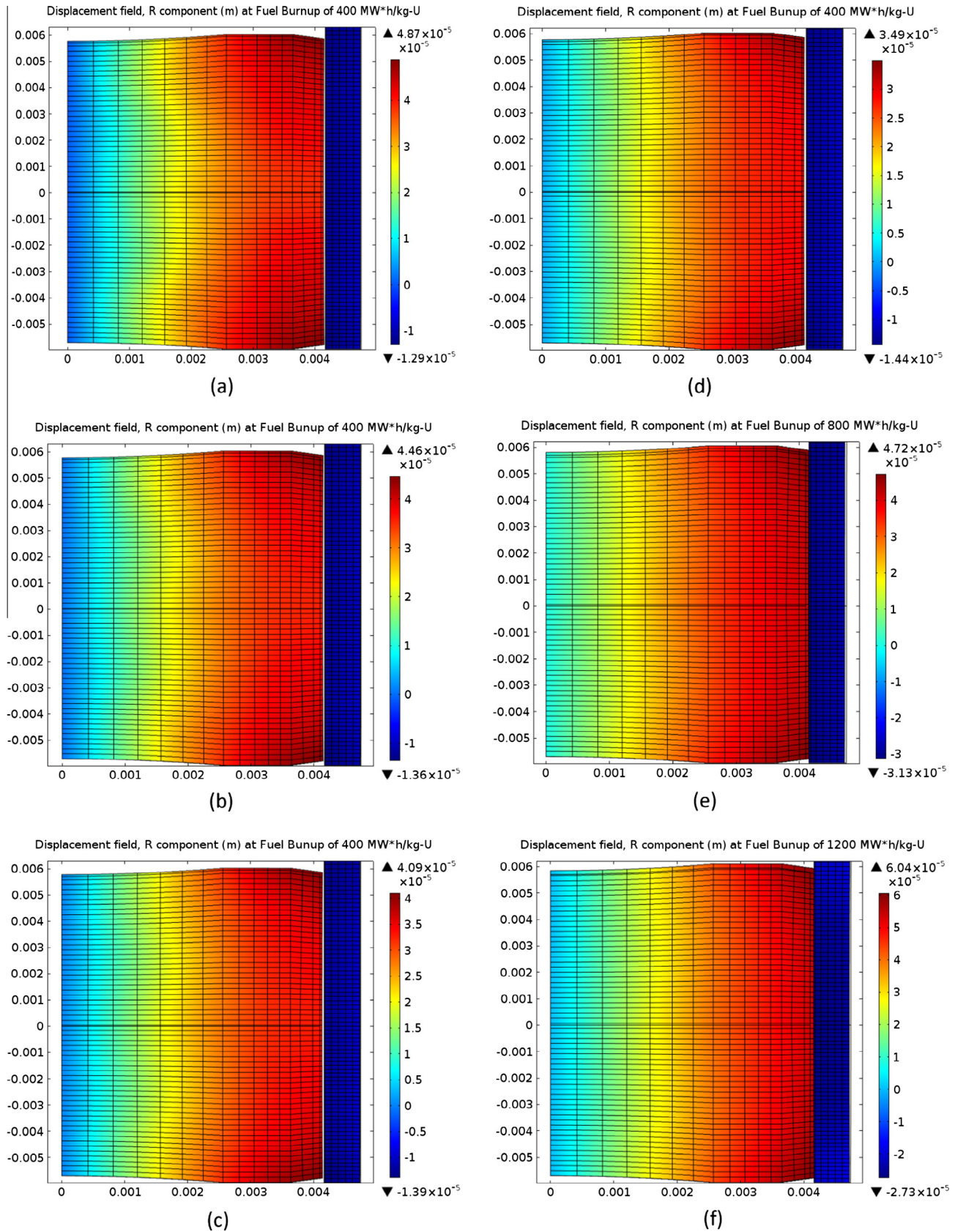


Fig. 14. Calculated fuel and cladding radial displacement with different type of fuels and Zircaloy cladding at different fuel burnup, all the distances are in unit meter. (a) UO_2 displacement at 400 MW* $h/kg-U$; (b) UO_2 -5 vol% SiC displacement at 400 MW $h/kg-U$; (c) UO_2 -10 vol% SiC displacement at 400 MW $h/kg-U$; (d) UO_2 -20 vol% SiC displacement at 400 MW $h/kg-U$; (e) UO_2 -20 vol% SiC displacement at 800 MW $h/kg-U$; and (f) UO_2 -20 vol% SiC displacement at 1200 MW $h/kg-U$.

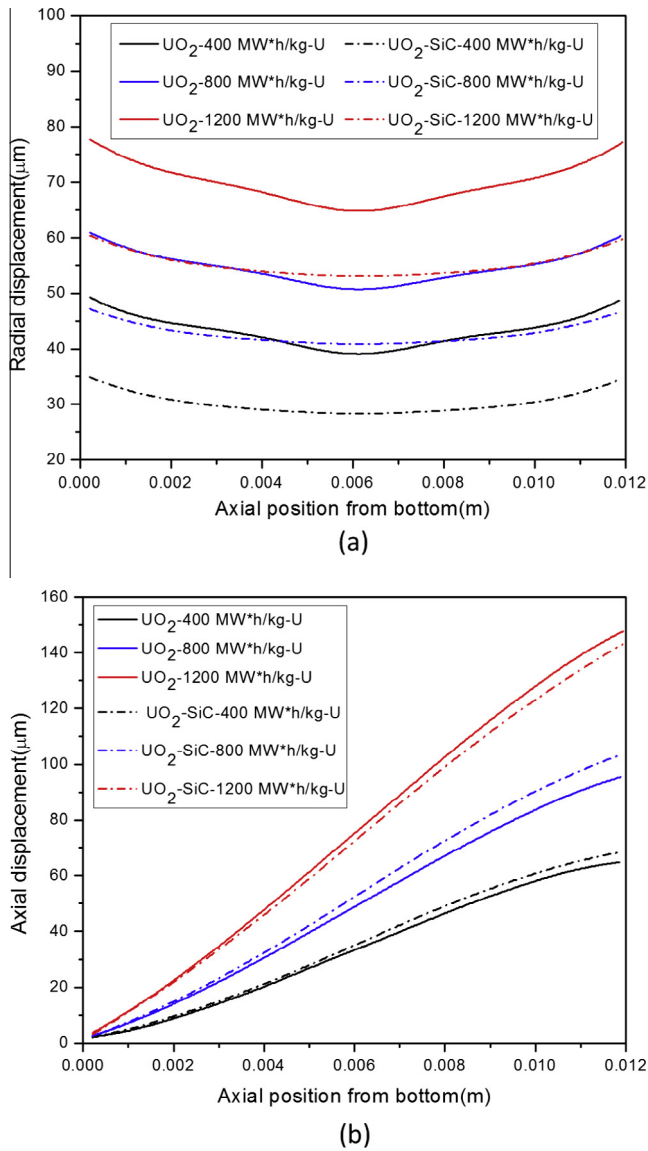


Fig. 15. Calculated radial (a) and axial (b) displacement over the UO₂ and UO₂-20 vol%SiC fuel surface at different fuel burnup.

Table 2

Input parameters for the axisymmetric problem.

Linear average power (W/cm)	200
Fast neutron flux (n/m ² s)	9.5×10^{17}
Coolant pressure (MPa)	15.5
Coolant temperature (K)	530
Coolant convection coefficient (W/m ² K)	7500
Rod fill gas	Helium
Fill gas initial pressure (MPa)	2.0
Initial fuel density	95% theoretical

determined by the fuel outer surface and cladding inner surface temperatures, thermal conductivities of fuel, gas and cladding, contact pressure, surface roughness conditions, gap size, etc. So from Fig. 9(a) and (b), the gas conductance was found to dominate the change of the total gap conductance, the gas conductance was found to increase rapidly due to a significant temperature increase of the fuel outer surface and cladding inner surface (as we can see from the inset of Fig. 10), then gradually increase due to the decrease of the gap size, and further increase dramatically just

before the gap closure. After the gap closure, the gas conductance was found to decrease due to the degradation of the gas thermal conductivity. Interestingly, before gap closure, the gas conductance of UO₂ fuel case was found to be the highest and UO₂-20 vol%SiC fuel case the lowest, while after gap closure, the trend was found to be opposite. This is because before gap closure the gap size dominates the value of gas conductance while after gap closure the gas thermal conductivity dominates the gas conductance (which is lower for pure UO₂ fuel due to fission gas release). The trend for the gap conductance in the four cases was observed to be similar with only small variations in timing and magnitude.

The fission gas release is depicted against fuel burnup in Fig. 11 for the four different cases. From Fig. 11, the UO₂ case experienced the highest fission gas release fraction as expected due to its low thermal conductivity and high fuel temperature. The fission gas release can be reduced by about 25% at the fuel burnup of about 1200 MW h/kg-U for UO₂-20 vol%SiC fuel. For UO₂-5 vol%SiC and UO₂-10 vol%SiC fuel, the fission gas release can be reduced by around 10% and 17%, respectively, at the fuel burnup of about 1200 MW h/kg-U. Fig. 11 also indicates that the earliest start time to release fission gas is found to be the case of UO₂ fuel without SiC addition. This is an expected observation as fission gas release is dominated by fuel temperature.

Fig. 12 shows the gap/plenum pressure evolution against fuel burnup for the four different cases (inset shows the evolution within the first three hours). Regular UO₂ fuel illustrates the highest gap/plenum pressure since its lower thermal conductivity leads to higher fuel temperature and more fission gas release. On the other hand, UO₂-20 vol%SiC fuel and Zircaloy cladding experiences lowest gap/plenum pressure as shown in Fig. 12.

Fig. 13 shows the evolution of the oxygen deviation from the stoichiometry, which is designated as x :

$$x = O/U - 2 \quad (70)$$

For the light water reactor, the fuel is in the hyperstoichiometric state, i.e., $x > 0$. The oxygen redistribution is more significant in the UO₂ fuel compared to the UO₂-SiC composite fuels due to the relatively high fuel temperature of the UO₂ fuel.

The radial displacements are shown in Fig. 14(a)–(d). The fuel radial displacements decrease with the SiC volume fraction increase from 0 to 20%, which is due to the fuel temperature decrease, at a certain burnup. Fig. 14(d)–(f) shows the fuel radial displacements increase with the fuel burnup increase at a fixed SiC volumetric fraction in the composite fuel. The radial and axial displacements are also depicted over the UO₂ and UO₂-20 vol% SiC fuel surface at different fuel burnup, as shown in Fig. 15(a) and (b). The radial displacement of UO₂ fuel is much larger than UO₂-20 vol%SiC fuel, although both fuels types take on a so-called “hourglass” shape (i.e., the radial displacements at the fuel top and bottom are much larger than at the middle part of the fuel, which is caused by the elastic stress induced by the gradient in temperature field) while the fuel axial displacement is found to have significant difference.

5. Conclusions

In summary, the enhanced thermal conductivity and improved fuel performance in UO₂ fuel mixed with various compositions of SiC has been modeled in a light water reactor using CAMPUS. Constitutive laws currently implemented include UO₂-SiC temperature and burnup dependent thermal properties, solid and gaseous fission product swelling, fuel densification, fuel thermal and irradiation creep, fission gas generation, diffusion and release, cladding thermal and irradiation creep, fuel-clad gap mechanics (contacts

and gap heat transfer), and gap/plenum gas behavior during irradiation. By developing the model and analyzing the model predictions a more complete understanding of the possible impact of $\text{UO}_2\text{-SiC}$ composite fuel on light water reactor fuel performance is presented in this paper.

SiC has unique and superior properties such as high thermal conductivity, low neutron cross section, high melting point, and great chemical stability, and was chosen as the secondary phase in the UO_2 matrix to enhance the thermal conductivity of the fuels. In our modeling and simulation work, the $\text{UO}_2\text{-SiC}$ enhanced thermal conductivity composite fuel showed the capability of decreasing the fuel centerline temperature by up to 150 K (for 20% SiC), increasing the gap total conductance, reducing the fission gas release, decreasing the gap/plenum pressure, lessening oxygen redistribution, and facilitating a reduction in pellet cladding interaction through lessening the thermal stresses that result in fuel cracking, relocation and swelling. The safety of the reactor operation would be then improved.

The $\text{UO}_2\text{-SiC}$ fuel performance model developed in this work is intended as an exploratory assessment of the impact of SiC additions to enhance UO_2 fuel performance. The phenomena models are based on the limited availability of the composite material property data extrapolated from unirradiated samples and single component behavior. Further work, including comparisons with experimental measurements from irradiated fuel samples, is required to refine and eventually validate the phenomena models. This model may serve as a stepping stone toward a more complete understanding of the $\text{UO}_2\text{-SiC}$ system.

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